SCS 3204: AUTOMATA THEORY

Lecture IV: Regular Expressions

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Recap

- A precise definition aids in carrying out proofs
- A concise definition assists in understanding
- Recursive definitions are precise and concise
- Regular expressions ➔ another method of defining languages precisely and concisely

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Need for more precision!

- We have used the following symbols to define languages e.g.

  L_1={x^n for n=1 2 3 ...}

  L_{ODD}={x^n for n=1 3 5 7...}

  L_5={x^n for n=1 4 9 16 ...}

  L_6={x^n for n=3 7 8 29 ...} ?
- For computation, more precise language-defining symbolism is required.

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Applying a closure directly to alphabet symbols

Consider L_4

L_4 = {Λ x xx xxx xxxx ...}

possible (previous) definitions:

- As a closure of a smaller set:

  Let S = {x}. Then L_4 = S*.

  shorthand representation: L_4 = {x}*
- Now, apply kleene star directly to *x* i.e.

  X*

  this simple expression, ***x**** indicates some sequences of *x*'s (or none at all).

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Applying a closure directly to alphabet symbols

- This notation is helpful in defining languages:

  L_4 = Language(**x***).

  since **x*** is any string of *x*'s, then L_4 is the set of all possible strings of *x*'s of any length (including Λ).
- Note:

  x* is a language-defining symbol, while L_4 is the name we give to a certain language.

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Example 1

Let us define language *L* over the alphabet Σ = {a b}

L = {a ab abb abbb abbbb abbbbb …}

What is the descriptive definition of *L*?

L = language(ab*)

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Example 2

L_1 = {Λ ab abab ababab ...}

What is the descriptive definition of L_1?

L_1 = *Language*(ab)*

Note!

ab* ≠ (ab)*

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Example 3

Σ = {x}

L_1 = {x^n *for n = 1 2 3 …*}

- How would you define the language L_1 using this notation?

  Remember that x^* generates Λ!

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Example 3 Solutions

L_1 = language(**xx***)

L_1 = language($\mathbf{x}^+$)

L_1 = language(**xx*x***)

L_1 = language(**x*xx***)

L_1 = language($\mathbf{x}^+$**x***)

L_1 = language(**x***$\mathbf{x}^+$)

L_1 = language(**x*x*x*xx***)

...

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Some more languages

Σ = {a b}

- ab*a
- a*b*
- a*b*a*
- What is the difference between the language denoted by abb*a and ab*a?
- Given Σ = {x} what are some alternative ways that you might define the Language L_{odd} using this notation?

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The plus sign (+) = Either .. or

- Consider the following language *T* defined over the alphabet Σ = {a b c}

  T = {a c ab cb abb cbb abbb cbbb abbbb cbbbb…}

- *What is the descriptive definition of T?*

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The plus sign (+) = Either .. or

- Consider the following language *T* defined over the alphabet Σ = {a b c}

  T = {a c ab cb abb cbb abbb cbbb abbbb cbbbb…}

- *What is the descriptive definition of T?*
- All the words in *T* begin with either an ***a*** or a ***c*** followed by any number of ***b***'s (including no ***b***'s).
- We can denote this language by using the following regular expression:

  T = language((a + c)b^*)

- The set of all strings that can be produced using this expression is the language of the expression.
- *How can you define T recursively?*

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Defining finite languages

- L = {aaa aab aba abb baa bab bba bbb} *i.e. all strings over the alphabet {a b} of length 3.*
- We can define this language in different ways using regular expressions
- L = language ((a + b)(a + b)(a + b))
- A shorthand way of expressing the same thing is:

  L = language $((a + b)^3)$
- A whole class of languages of strings of a's and b's of various lengths can be defined using the regular expression L = language $((a + b)^n)$
- Finally the infinite language that contains all possible strings of a's and b's of finite length can be defined using the regular expression $(a + b)^*$

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How could we define the following languages?

Given Σ = {a b},

- How do you define the language of all words that begin with the letter ***b*** followed by ***a***'s and ***b***'s of any length?
- How do you define the language of all words that begin with an ***a***, then anything (any combination of ***a***'s and ***b***'s) and ends with a ***b***?
- The language of all strings that have at least one ***b*** in them?
- The language of all strings of b's?

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Formal Definition of Regular Expressions

- The languages defined by regular expressions are referred to as regular languages.
- A regular language is one that can be defined by a regular expression even though it may also have many other fine definitions.
- Regular expressions, though powerful and concise, are of limited capacity since there are many languages that cannot be defined by regular expressions.
- The ONLY symbols that appear in regular expressions are:
  1. The letters of the alphabet Σ,
  2. Λ,
  3. Parentheses,
  4. Kleene star operator (*) and
  5. The Either..Or operator (+).

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Recursive definition of Regular Expressions

Rule 1: Every letter of Σ is a regular expression. Λ itself is a regular expression.

Rule 2: If r_1 and r_2 are regular expressions then so are (r_1) r_1r_2 $r_1 + r_2$ r_1^*

Rule 3: Nothing else is a regular expression (exclusion rule)

why don't we have r_1^+ as part of the definition?

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Union of Regular Expressions

- Consider the language with alphabet Σ = {a b}

  which is defined by the regular expression:

  $(a + b)^*a(a + b)^*$
- What is the descriptive definition of this language?
- Is *abbaab* a string in this language?
- What regular expression defines those words omitted from this language?

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Union of Regular Expressions

- If we combine the two regular expressions, we should produce the language of all strings, since:

  (all strings with an *a*) + (all strings without an *a*) = all strings

  $(a + b)^*a(a + b)^* + b^* = (a + b)^*$

Here, we have joined two language-defining expressions to produce an expression that defines the union of the two languages defined by the individual expressions.

Note! Not similar to addition in the algebraic sense, since

$a^* = a^* + a^*$
$a^* = a^* + a^* + a^*$
$a^* = a^* + aaa$
...

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More language definitions

$\Sigma = \{a\ b\}$

- What about the language with exactly three *a*'s?
- What about the language with at least one *a* and at least one *b*?
  - What is another way that we could define this language?
- Using the regular expressions defined above, how can you define all possible strings of *a*'s and *b*'s i.e. $(a + b)^*$?

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More on equivalence

$(a + b)^* = (a + b)^* + (a + b)^*$

$(a + b)^* = (a + b)^*(a + b)^*$

$(a + b)^* = \Lambda + a(a + b)^* + b(a + b)^*$

$(a + b)^* = (a + b)^* ab(a + b)^* + b^* a^*$

Note! These are not algebraic polynomials!

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Distributive Law

$\Sigma = \{a\ b\}$

Let $V = \{\Lambda\ a\ b\ ab\ bb\ abb\ bbb\ abbb\ bbbb\ ...\}$

What is the descriptive definition of this language?

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Distributive Law

$\Sigma = \{a\ b\}$
Let $V = \{\Lambda\ a\ b\ ab\ bb\ abb\ bbb\ abbb\ bbbb\ ...\}$

What is the descriptive definition of this language?

The regular expression defining V is:
$b^* + ab^*$

Alternatively, V can be defined by $(\Lambda + a)b^*$

since $b^* = \Lambda b^*$, applying distributive law we get:

$\Lambda b^* + ab^* = (\Lambda + a)b^*$

Here, we have factored out b^* by taking advantage of the distributive law.

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Distributive Law

Let $T = \{a\ c\ ab\ cb\ abb\ cbb\ ...\}$

What is the descriptive definition of this language?

What alternate regular expressions define this language?

Note: Distributive law must be used with care since expressions may be distributed but operators cannot!

e.g. $(ab)^* \neq a^*b^*$

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The Product Set of two Languages

Definition:

- If S and T are two sets of strings of letters (whether they are finite or infinite sets) we define the product set of strings of letters to be:

  ST = {all combinations of a string from S concatenated with a string from T in that order}

Example:

S = {a aa aaa} T = {bb bbb}, then

ST = {abb abbb aabb aabbb aaabb aaabbb}

(Note that these words are not in lexicographic order, but in size ordering!)

If S = {aba bab} and T = {a b} what is the product set ST?

If P = {a bb bab} and Q = {Λ bbbb} what is the product set PQ?

Note: if L is any language, then $L\Lambda = \Lambda L = L$

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An important definition

- The rules for associating a language with a regular expression can be rigorously defined using a recursive definition:
- Rule 1: The language associated with Λ is just $\{\Lambda\}$, a one-word language.
- Rule 2: The language that is associated with a single letter regular expression is that one letter word alone.
- Rule 3: If r_1 is a regular expression associated with the language L_1 and r_2 is a regular expression associated with the language L_2, then:

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An important definition

- The regular expression $(r_1)(r_2)$ is associated with the product L_1L_2:

  $$\text{language}(r_1r_2) = L_1L_2$$

- The regular expression $r_1 + r_2$ is associated with the language formed by the union of sets L_1 and L_2:

  $$\text{language}(r_1 + r_2) = L_1 + L_2$$

- The regular expression $(r_1)^*$ is assocated with the language L_1^*, the kleene closure of set L_1:

  $$\text{language}(r_1)^* = L_1^*$$

Rule 4: Exclusion rule

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Recap

- Recursive definition proves that there is some language associated with every regular expression.
- As we build up a regular expression from the rules, we simultaneously build up the corresponding language.
- How do we determine if two regular expressions define the same language?
- Every regular expression is associated with some language; is it also true that every language can be described by a regular expression?

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Finite Languages are Regular

Theorem: If L is a finite language (i.e. has finitely many words), then L can be defined by a regular expression.

Proof:

if L = {baa abbba baba} then baa + abbba + baba is the regular expression that defines it.

If T = {aa ab ba bb}, then using the algorithm above, the regular expression that defines T is aa + ab + ba + bb.

What is another regular expression that defines T?

Regular expressions for a language need not be unique, we need only to show that one exists!

The algorithm above does not work for infinite languages. Why?

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Finite languages are regular

Using the theorem, what is the regular expression for L, where L = {Λ x xx xxx xxxx}?

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Cryptic Regular Expressions

- Consider the following languages defined over the alphabet Σ = {a b}:
  - What regular expression defines the language of strings of a's and b's that at some point contain a double letter?
  - How can you expand this expression to be one that defines all strings?
- Clearly, the result is not obvious!

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Another Example

- Consider the language EVEN-EVEN with $\Sigma = \{a\ b\}$ whose strings have an even number of a's and an even number of b's
- What is the regular expression that defines E?

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Exercises

- Construct a regular expression defining the following languages over the alphabet {0 1}.
  1. All words in which 0 appears quadrupled, if at all. This means that every clump of 0's contains 4 or 8 or 12 or 16 .... 0's.
  2. All words that contain exactly two 1's or exactly three 0's, not more.
  3. All strings that end in a double letter.
- Give regular expressions for the following over the alphabet $\Sigma = \{a, b, c\}$:
  1. All strings containing exactly one ***a***.
  2. All strings containing atleast two ***a***'s
  3. All strings containing at least one occurrence of each symbol in Σ.

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Exercises

1. Provide a regular expression for the following language defined over the alphabet $\Sigma = \{a,b\}$.
   $L_1 = \{b^x a^y : x \geq 1, y \geq 1, x+y \geq 3\}$
2. Give a recursive definition for the language L defined over the alphabet $\Sigma = \{a,b,c,d\}$, whose words are described by the regular expression: (da + cb)* (a + bb)
3. What is the regular expression defining the language defined over the alphabet $\Sigma = \{a,b,c\}$ whose strings do not contain substrings of a's of length greater than four?

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Exercises

1. Let $L = \{ab, aa, baa\}$. Which of the following strings are in L^* ?
   i. abaabaaabaa
   ii. aaaabaaaa
   iii. baaaaabaaaab
   iv. baaaaabaa
2. Find the regular expression defining
   $L = \{ba^n Q : n \geq 4, Q \in \{a,b\}^*\}$
3. Give a recursive definition for the language L defined over the alphabet $\Sigma = \{a,b,c,d\}$, whose words are described by the regular expression $ab(b(c+d))^*$.
4. Find the regular expression defining
   $L = \{a, bb, aa, abb, ba, bbb, \ldots\}$

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